Code No: 135BY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
COMPUTER GRAPHICS
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used Papers March-2024

PART – A

(25 Marks)

1.a)	Distinguish between pixel ratio and aspect ratio.	[2]
b)	Write an algorithm of Boundary fill.	[3]
c)	What is composite transformation?	[2]
d)	Explain 2 D translation with diagrams.	[3]
e)	List the properties of Bezier Curve.	[2]
f)	Explain the 3D translation.	[3]
g)	What are the limitations of Back face detection?	[2]
h)	Differentiate between Gourand shading and Phong shading.	[3]
i)	What is morphing?	[2]
j)	What is raster animation?	[3]

PART – B

(50 Marks)

2.a) Write an algorithm of Bressenham's Ellipse generation.
b) Explain about the Shadow mask method. [5+5]

OR

3.a) Consider the line from (5, 5) to (13, 9). Use the Bresenham algorithm to rasterize the line.
b) Explain about the working principle of Plasma Panel display device. [5+5]

4.a) Obtain the mirror reflection of the triangle formed by the vertices A(0, 3), B(2, 0) and C(3, 2) about the line passing through the points (1, 3) and (-1,-1).
b) Write about shear trans formation. [6+4]

OR

5.a) What do you understand by clipping? Give line clipping algorithm.
b) Explain about the cyrus-back line clipping algorithm. [5+5]

6.a) What are the problems that are encountered in perspective projections?
b) Derive the transformation matrix for rotation about the Y-axis for three dimensional object. [4+6]

OR

7.a) Write the steps involved in constructing 3D viewing transformation.
b) Find the transformation matrix for Oblique projection onto xy plane. [5+5]

8.a) Explain about the Flat shading model.
b) What are the advantages and disadvantages of octree method? [6+4]

OR

9.a) Explain about the Gouraud shading method.
b) Explain about the BSP – tree method. [5+5]

10. Describe about the computer animation languages. [10]

OR

11.a) What are the principles of computer animation?
b) Explain the design of animation sequences. [5+5]

---ooOoo---

Used Papers March-2024